

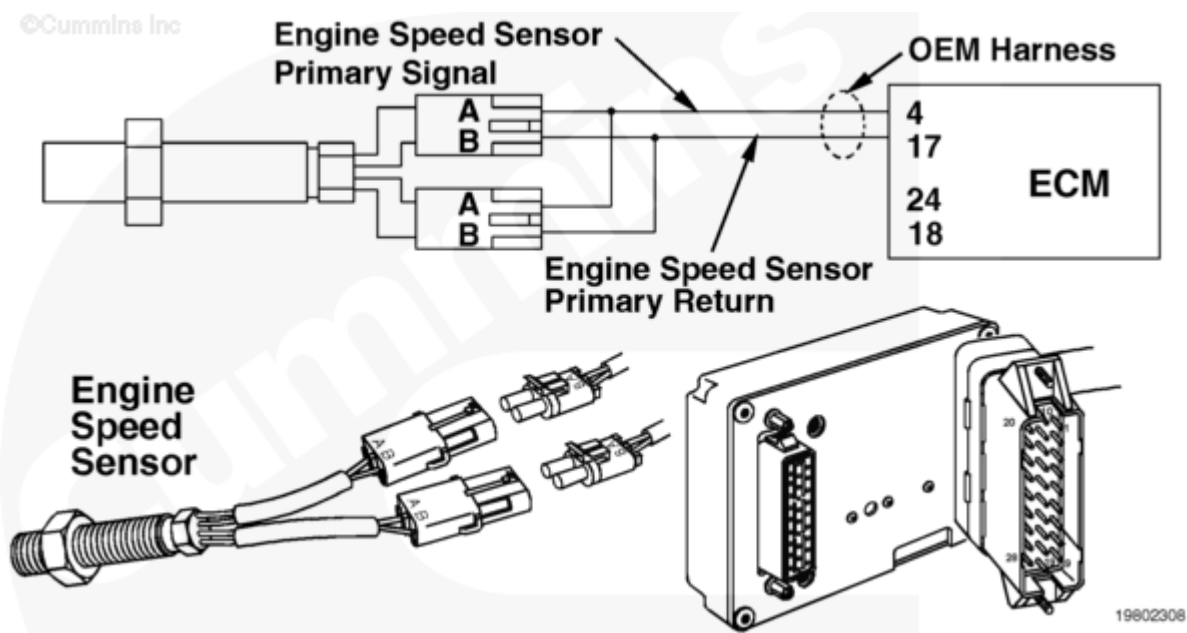
Fault Code: 115-fa

Engine Magnetic Speed/Position Lost Both of Two Signals - Data Erratic, Intermittent, or Incorrect

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 115-fa PID(P): 190 SPN: FMI: 2 Lamp: Flashing SRT: 00-621	Both signals failed to be detected at the engine position sensor circuit.	Derate power and engine can run at high idle only .



Engine Speed Sensor Circuit

Circuit Description

The engine speed sensor is dual channel speed sensor used by the ECM to monitor the engine speed. The speed sensor has four circuits: two signal circuits, and two return circuits. As the teeth on the crankshaft flywheel move past the speed sensor, a signal is generated on the speed sensor signal circuits.

Component Location

The engine speed sensor is mounted in the flywheel housing on the rear of the engine.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected a signal failure with engine speed sensor.

Action Taken When The Fault Code Is Active

- The ECM illuminates the red STOP ENGINE lamp immediately when the diagnostic runs and fails.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.

Conditions For Clearing The Fault Code

- Turn the keyswitch OFF. Allow the ECM to completely power down and turn the keyswitch ON.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the red STOP ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.

- Malfunction or damaged engine speed sensor.

Refer to Troubleshooting Fault Code t05-115 (</qs3/pubsys2/xml/en/procedures/98/98-t05-115fa.html>)

Last Modified: 15-Sep-2021
